

EchoSarathi — Voice AI for Bharat

A self-serve voice agent platform for Indian SMBs

Social Innovation and Entrepreneurship (A500507) · Review-II

[College Name] · [Semester, Academic Year]

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Presentation Outline

- Problem Statement
- Existing Solutions
- Gaps in Existing Solutions
- Proposed Solution
- Components Required
- Block Diagram
- Flow Chart
- Working Principle of Components
- Working of the Model
- Results and Conclusions
- References

Problem Statement — 60M Indian SMBs, no affordable voice calling

- **Manual and expensive:** schools call parents one by one; clinics remind patients by hand; shops confirm orders over phone. Staff time lost — or the calls never happen.
- **Enterprise voice AI is unaffordable:** Ringg/Gnani/Uniphore charge Rs. 6+/min with enterprise contracts. A school in Warangal cannot buy that.
- **SMB tools are black boxes:** no transcript, no latency, no audit of why a call failed. Owners cannot improve what they cannot see.

Existing Solutions

- **Ringg AI** — enterprise voice agents (Practo, Flipkart); \$15.5M funding; Rs. 6/min. SMBs excluded by contracts.
- **Outpero** — SMB voice agent, Rs. 4,999/mo + Rs. 3.5/min, 10+ languages. Black box: no stack visibility.
- **ThinnestAI** — Rs. 1.5–2/min + passthrough. Cheapest, but developer-only: no templates, no dashboard.
- **Human BPO agents** — Rs. 25–40/min, 60–80% attrition. Expensive, inconsistent, unscalable.

Gaps in Existing Solutions

Gap	Who has it	What's missing
SMB affordability	Ringg, Gnani, Uniphore	Enterprise pricing blocks 60M+ SMBs
Vertical templates	ThinnestAI, Vapi, Retell	Require code; no pre-built agents
Observability	All SMB platforms	No per-turn latency / field audit
Vertical depth	Vacademy (edu-only)	Locked to one vertical
Code-switching depth	Most others	Coverage ≠ depth; mid-sentence Telugu-English rare
Data sovereignty	Global platforms	Voice data leaves India; DPDP risk

No platform combines **SMB pricing + vertical templates + observability + data sovereignty**.

Proposed Solution — EchoSarathi

- Sign up → pick a template (attendance, appointment, fee, COD) → fill contact card → test in browser playground → launch campaign.

Feature	EchoSarathi	Outpero	ThinnestAI	Ringg
Self-serve	Yes	Yes	Yes	No
Vertical templates	Yes	Partial	No	Enterprise
Per-turn latency visible	Yes	No	No	No
Extracted fields audit	Yes	No	No	No
Data sovereignty	Self-hosted	No	No	No
Price per minute	Rs 2.99	Rs 3.50	Rs 2.14	Rs 6.00

Pricing: ■4,999/mo (500 min included) + ■2.99/min overage.

Components Required

Hardware

Item	Purpose
Cloud VM (4 vCPU, 8GB)	Hosts LiveKit, backend, voice worker
Smartphone + Indian SIM	End-to-end test calls to +91 numbers
USB headset / mic	Browser playground testing
Developer laptop	Development, testing, deployment

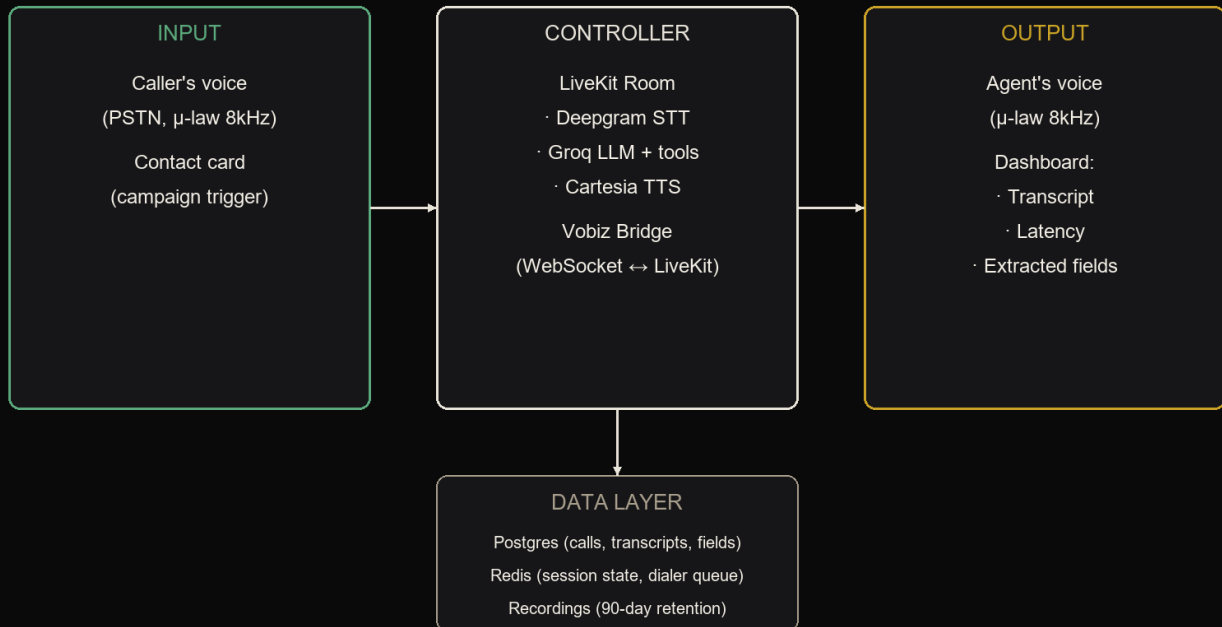
Software

Layer	Technology	Purpose
Orchestration	LiveKit (self-hosted)	Rooms, agent lifecycle, audio routing
Telephony	Vobiz	India PSTN, mu-law 8kHz WebSocket
STT	Deepgram Nova-2 Phonecall	Streaming ASR, telephony-tuned
LLM	Groq Llama 3.1 8B Instant	Responses + tool calling, <200ms TTFT
TTS	Cartesia Sonic / Sarvam Bulbul V3	Streaming speech, 8kHz out
Backend	Python FastAPI	Calls, campaigns, webhooks
Frontend	React	Dashboard, Call Detail, playground
Database	SQLite / Postgres	Calls, transcripts, fields
Cache	Redis	Session state, dialer queue
Deploy	Docker Compose	Containers

Block Diagram

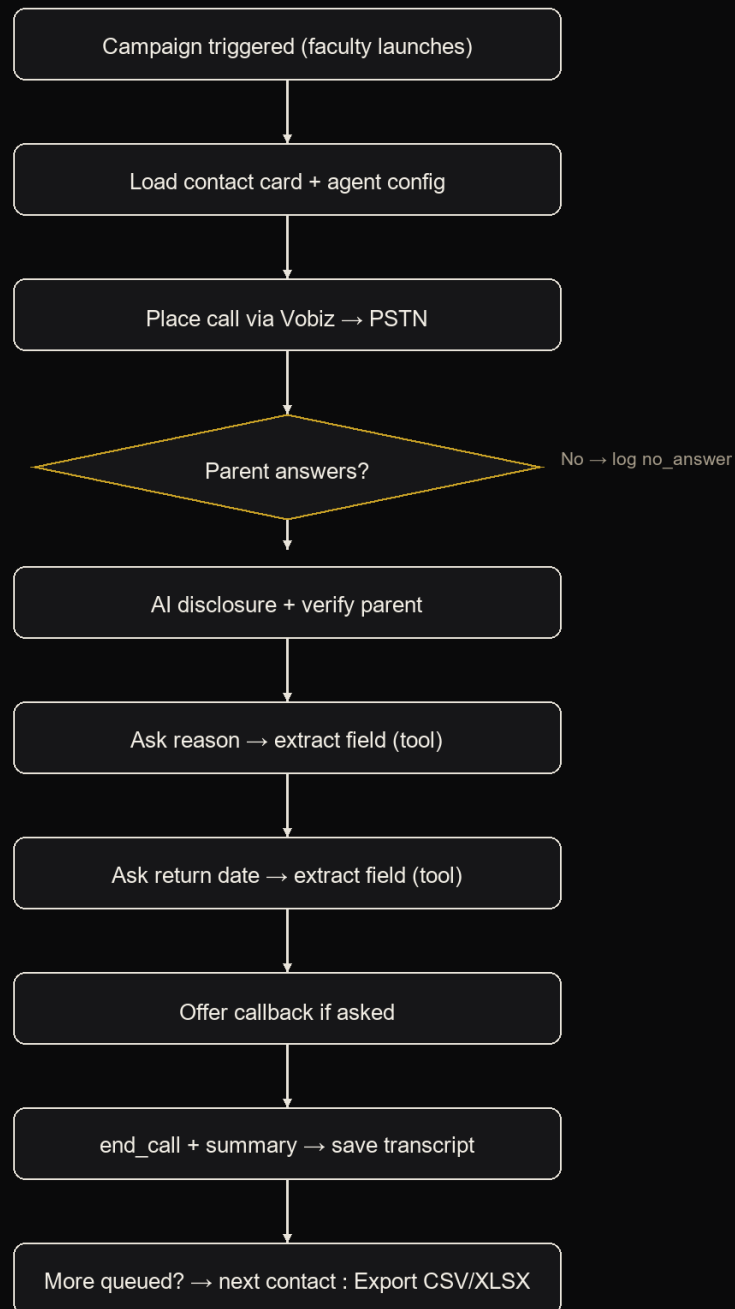
EchoSarathi — System Block Diagram

INPUT (sensors) → CONTROLLER → OUTPUT (monitor)



Flow Chart

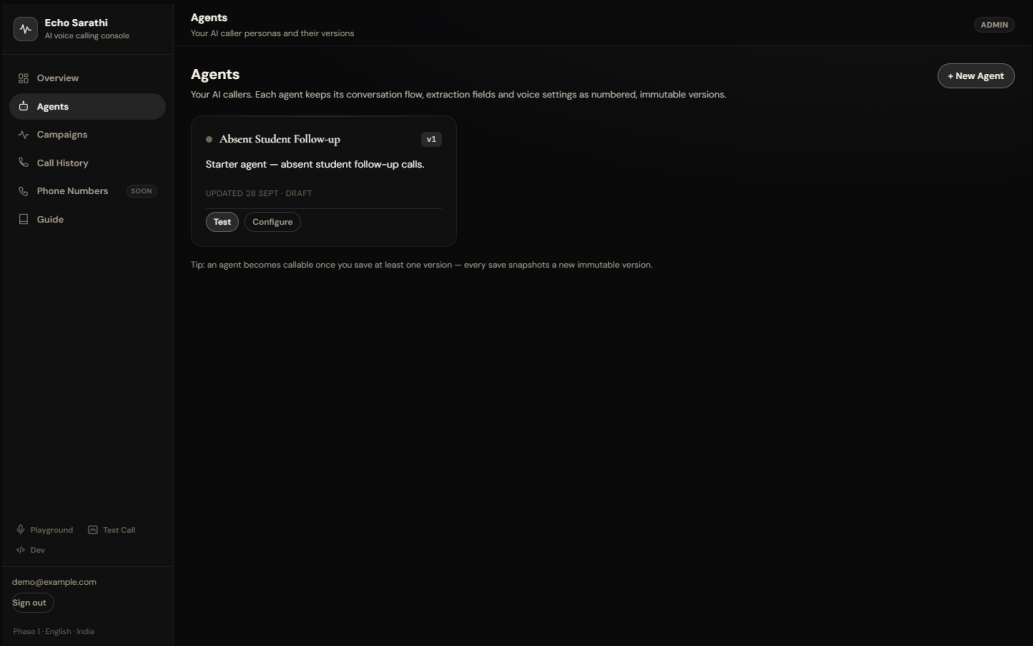
EchoSarathi — Call Flow Chart



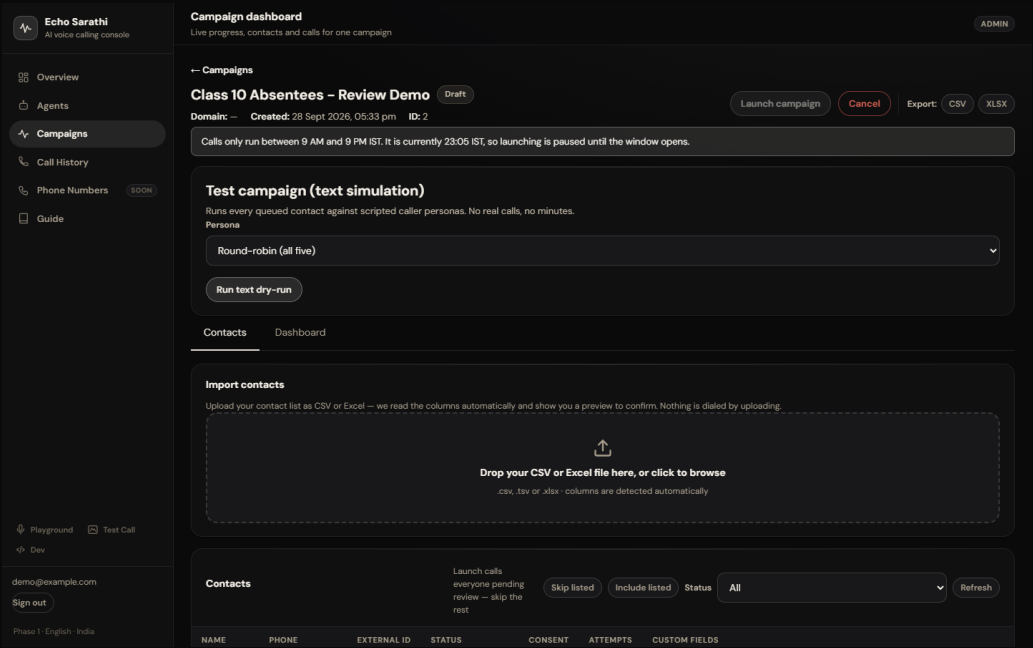
Working Principle of All Components

- **Vobiz:** dumb pipe — PSTN audio (mu-law 8kHz) to WebSocket and back.
- **LiveKit:** one room per call; routes audio bridge ↔ worker; turn-taking + barge-in.
- **Deepgram:** interim + final transcripts; endpointing 300ms.
- **Groq LLM:** replies + tool calls (record fields, end call); 300-token cap.
- **Cartesia:** streams first audio in ~150ms, sentence by sentence.
- **Backend/Frontend/Data:** FastAPI control + React dashboard; Postgres records, Redis queue, 90-day recordings.

Project Model — Agents & Campaign

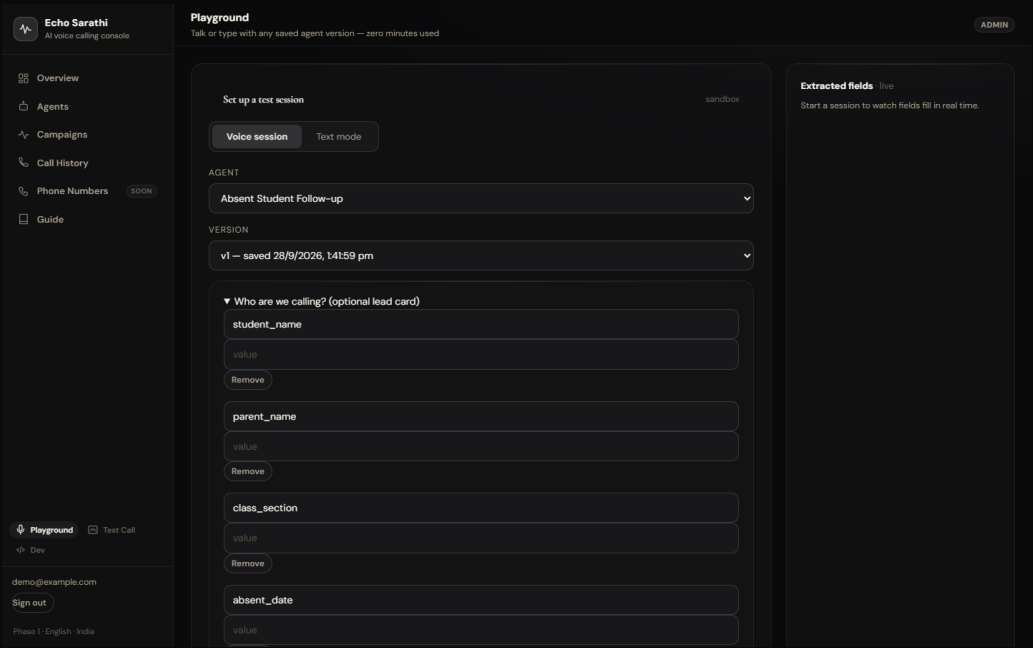


Agent workspace — persona, versions, voice settings.

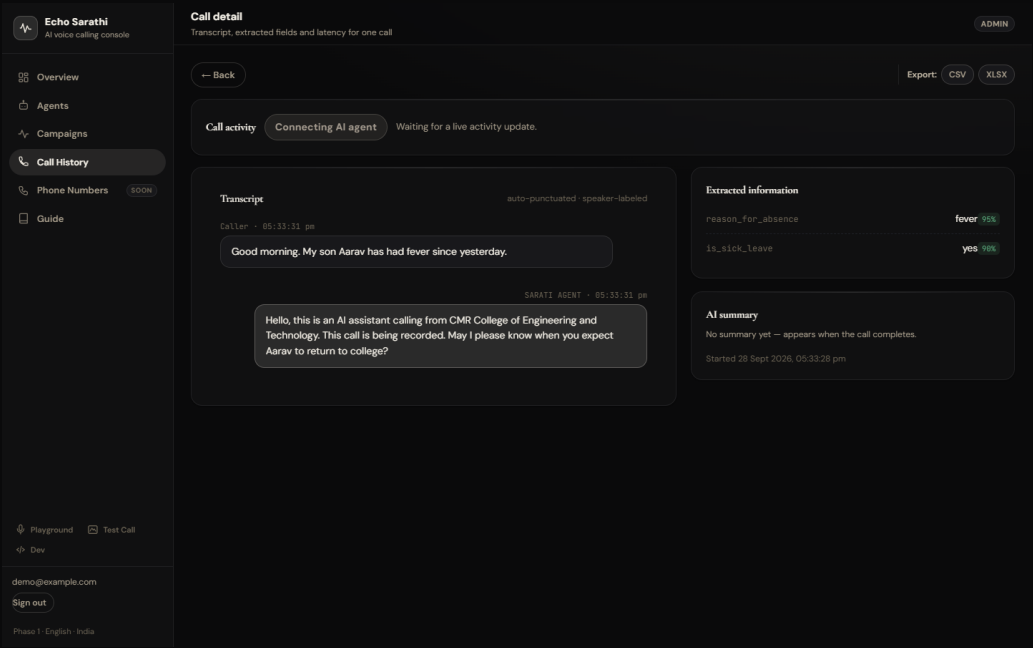


Campaign roster — contacts with Skip/Include selection.

Project Model — Playground & Call Detail



Browser playground — test any agent version, zero minutes.



Call Detail — transcript, extracted fields with confidence.

Results

Metric	Result
Real outbound calls	Working to +91 numbers via Vobiz
TTS first audio	117ms
Greeting	~35 words / under 10s
Cost per 50s call	Rs 0.80
Agent templates	2: attendance follow-up, appointment reminder
Contact card flow	Verified end-to-end
Tests	Backend 213/213 · voice-agent 110/110

Ongoing: turn-detection tuning, greeting pre-render, interruption handling, latency-consistent model.

Conclusions

- Voice AI for Indian SMBs is viable at Rs. 2.99/min with 55–66% margin.
- The gap (SMB pricing + templates + observability + sovereignty) is real and fillable.
- Observability — per-turn latency, fields, outcomes in one view — is the differentiator.
- Self-hosted data path answers DPDP objections from hospitals/NBFCs.
- Code-switching (Sarvam, Phase 2) opens Tier-2/3 India.
- Hard engineering done; remaining: self-serve layer, DLT, billing.

References

- IEEE Access 2023 — voice UI design for Indian languages (IEEE Xplore).
- IEEE/ACM TASLP 2024 — end-to-end ASR for low-resource Indian languages.
- IEEE TNNLS 2024 — LLMs for conversational AI survey.
- ICASSP 2025 — latency optimization in cascaded STT-LLM-TTS pipelines.
- Industry: ringg.ai · outpero.com · thinnest.ai · vobiz.ai · docs.livekit.io/agents · sarvam.ai · DPDP Act (meity.gov.in).
- *Verify exact titles/years on IEEE Xplore before submission.*

THANK YOU

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Questions?

Contact: [team email] | [project repo link]